SOLUTION REQUIREMENTS SPECIFICATION

Developer: [Left Blank] | OptiCrop Project Documentation
Repository: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

1. Functional Requirements

- Soil Input Form: Users can input Nitrogen, Phosphorous, Potassium, Temperature, Humidity, Soil pH, and Rainfall.
- Soil Categorization: The system maps soil chemistry inputs to one of five soil diagnostic groups via K-Means clustering.
- Expected Yield Forecast: The system estimates crop yield (tons/hectare) using a Random Forest regressor.
- NPK Vitality Prescription Engine: Compares inputs against crop thresholds and outputs fertilizer guidance.
- Model Telemetry Dashboard: Displays accuracy comparison charts, confusion matrices, and feature importances.
- History Logging: Persists user inputs and system recommendations to a local CSV file for performance analysis.

2. Non-Functional Requirements

- Accuracy: The core crop recommendation classifier must achieve a classification accuracy greater than 98% on test data.
- Response Time: Predictions must render in the UI in less than 1.0 seconds.
- Interface: The UI must be fully responsive and support mobile devices.
- Portability: The system must run locally in a standard virtual environment.

1	Soil parameter input form (NPK, pH, rainfall, temperature, humidity)	Functional	High
2	Random Forest crop recommendation with 100% test accuracy	Functional	High
3	Custom NPK correction: prescribe Urea / SSP / MOP amendments	Functional	High
4	Automated unit tests covering Flask routes and prediction bounds	Non-Functional	High